

Urban Ghost Kitchen

Data-Driven Business Insights Report

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September 2026

Scope: This report analyzes SQL-based query results covering Customer, Revenue, Order, Kitchen, Delivery, Menu/Category, Customer Segmentation, Cancellation/Refund, and Time-Based dimensions for the fictional ghost kitchen business *Urban Ghost Kitchen*. All figures are drawn directly from the provided query outputs; no numbers, trends, or causes are assumed beyond what the data supports.

1. Executive Summary

- Cancellations are heavily concentrated in a small number of kitchens.** Overall cancellation rate is 10.19% (2,038 of 20,000 orders), but four kitchens – KIT-017 (18.73%), KIT-009 (18.44%), KIT-001 (18.26%), and KIT-005 (16.93%) – run at roughly double the network average, while the next-highest kitchen (KIT-003) drops to 10.15%.
- Revenue is driven far more by price per item than by sales volume.** Order quantity across the 10 menu categories is tightly clustered (8,706–9,124 units, a ~5% spread), but category revenue ranges from ₹837,752 (Beverage) to ₹3,248,289 (Pizza) – a ~4x spread – explained almost entirely by average unit price (₹95.79 for Beverage vs ₹363.34 for Pizza).
- One kitchen (KIT-002) dominates order volume and revenue.** It handled 3,427 orders (17.1% of all orders) and ₹1,949,309.57 in revenue (16.9% of total revenue) – more than double the second-largest kitchen (KIT-025: 1,349 orders, ₹786,527.51).
- Short-distance deliveries cancel more often than long-distance ones.** Cancellation rate is 12.10% for “Short” distance orders vs 7.53% for “Long” and 6.69% for “Medium” – the opposite of what delivery-distance risk would typically predict.
- Premium customers show a split pattern: lowest cancellation, highest refund rate.** Premium cancellation rate (10.20%) sits between New (11.14%) and Regular (9.92%), but Premium has the highest refund rate at 4.52% vs 3.99% (Regular) and 3.23% (New).
- Peak demand is a tight, predictable evening window.** 18:00–21:00 accounts for 6,742 of 20,000 orders (33.7%), peaking at 20:00 with 2,215 orders – 22x the 3:00–4:00 AM trough (98–99 orders).
- Revenue is not customer-concentrated.** The single largest customer (CUS-00582, 276 orders) contributes only 1.36% of total revenue, and no customer in the top 20 exceeds that – indicating no single-customer dependency risk in the current data.
- Kitchens with the worst cancellation rates and kitchens with the worst refund rates are largely different kitchens.** Top cancellation offenders (KIT-017, KIT-009, KIT-001, KIT-005) do not overlap with top refund offenders (KIT-019: 5.99%, KIT-010: 5.81%, KIT-008: 5.52%, KIT-013: 5.04%), suggesting these are two separate operational problems, not one.

2. Detailed Business Insights

Area	Key Finding	Evidence	Business Implication
Customer	Revenue base is not concentrated in a few customers	Top customer CUS-00582 contributes 1.36% of ₹11,507,673.68 total revenue; top 20 customers each contribute under 1.4%	Low dependency risk on any single customer; revenue stability doesn't hinge on retaining one account
Revenue	Total revenue is ₹11,507,673.68, but 14.3% of it never converts to delivered revenue	Revenue by status: Delivered ₹9,866,018.28 (85.7%), Cancelled ₹1,181,250.94 (10.3%), Refunded ₹460,404.46 (4.0%)	Roughly ₹1.64M in booked order value is lost annually to cancellations/refunds – a direct leakage figure
Orders	Order status splits to 85.81% Delivered, 10.19% Cancelled, 4.00% Refunded	Orders by Status: Delivered 17,162 / Cancelled 2,038 / Refunded 800 of 20,000	The delivered rate is the baseline to track; a >10% cancellation rate is a meaningful chunk of operational capacity spent on orders that don't complete
Kitchen	Order and revenue volume are highly uneven across kitchens	KIT-002: 3,427 orders / ₹1,949,309.57 revenue vs KIT-021: 342 orders (lowest tier) – a ~10x spread in order count across 25 kitchens	Capacity planning and quality control need to be kitchen-specific, not network-wide, given this size disparity
Delivery	Delivery time scales with distance category, but two kitchens are outliers regardless of that pattern	Avg delivery time: Short 20.45 min, Medium 34.49 min, Long 57.81 min; but KIT-025 averages 41.50 min and KIT-009 averages 40.49 min overall, vs network average 29.31 min	These two kitchens likely have delivery-time issues independent of geography – worth investigating dispatch/staffing at KIT-025 and KIT-009 specifically
Menu/Categ	Revenue disparity across categories is driven by price, not by demand	Order quantity range is narrow (8,706–9,124 units across categories) but revenue ranges from ₹837,752 (Beverage) to ₹3,248,288.52 (Pizza), tracking avg unit price (₹95.79 vs ₹363.34)	Beverage and Dessert categories may be underpriced relative to their order volume, or Pizza's price point is what's carrying category-level revenue
Customer Segmentation	Segment revenue differences come from customer count and order frequency, not order value	AOV is nearly identical across segments (Premium ₹577.59, Regular ₹575.33, New ₹572.20 – under 1% spread), but avg customer spend differs sharply (Premium ₹19,720.55, Regular ₹12,278.32, New ₹6,597.31)	Segment value is a function of retention/frequency, not basket size – loyalty and repeat-order programs matter more than upsell here

Area	Key Finding	Evidence	Business Implication
Cancellation	New customers cancel most; Premium customers refund most	Cancellation rate: New 11.14%, Premium 10.20%, Regular 9.92%. Refund rate: Premium 4.52%, Regular 3.99%, New 3.23%	New customers may be dropping orders early (onboarding friction), while Premium customers complete orders but seek refunds afterward (possible quality/expectation gap)
Time-Based	Demand is concentrated in a narrow evening band and skews toward weekends	Orders by hour peak at 20:00 (2,215) and 19:00 (2,096); Saturday (3,338) and Sunday (3,251) are the two highest days of week, each above the 2,620–2,842 weekday range	Staffing and kitchen capacity should be weighted toward the 18:00–21:00 window and weekends specifically

3. Key Performance Highlights

- **Highest-performing customer:** CUS-00582 – 276 orders, ₹156,944.26 total spend (1.36% of total revenue), highest of any customer in the dataset.
- **Highest-revenue kitchen:** KIT-002 – ₹1,949,309.57 (16.9% of total revenue), also highest order volume (3,427 orders).
- **Highest-performing menu category (by revenue):** Pizza – ₹3,248,288.52, despite ranking only 4th of 10 in quantity sold (8,938 units).
- **Top menu item:** Insufficient evidence – the “Top 10 Menu Items” results are all Pizza SKUs (Cheese Burst, Pepperoni, BBQ Chicken, Farmhouse, Paneer Tikka, Margherita), but every item is tied at just 5 units sold each. This indicates no single item has meaningfully higher demand than others; the “top” list reflects the highest tier of a very fragmented, long-tail catalog rather than one standout product.
- **Peak ordering period:** 20:00 hour, 2,215 orders – the single busiest hour recorded; the broader 18:00–21:00 window accounts for 33.7% of all daily orders.
- **Fastest delivery pattern:** KIT-011, 22.63 min average delivery time – fastest of all 25 kitchens.
- **Slowest delivery pattern:** KIT-025, 41.50 min average delivery time – slowest of all 25 kitchens, and notably also the network’s 3rd-highest order volume kitchen (1,349 orders).
- **Highest cancellation area:** KIT-017 at kitchen level (18.73%); New customer segment at segment level (11.14%); Short-distance orders at distance level (12.10%).
- **Highest refund area:** KIT-019 at kitchen level (5.99%); Premium customer segment at segment level (4.52%); Medium-distance orders at distance level (4.37%).

4. Operational Issues

Issue	Evidence	Possible Business Impact
Four kitchens run cancellation rates far above network average	KIT-017 (18.73%), KIT-009 (18.44%), KIT-001 (18.26%), KIT-005 (16.93%) vs overall 10.19%, with a sharp drop to 10.15% at the next kitchen	These four kitchens alone account for a disproportionate share of the network’s 2,038 cancelled orders; fixing them would move the overall rate materially

Issue	Evidence	Possible Business Impact
KIT-025 and KIT-009 have unusually slow delivery times	41.50 min and 40.49 min respectively vs network average of 29.31 min, and vs the fastest kitchen (KIT-011) at 22.63 min	Slower delivery on high-volume kitchens (KIT-025 is 3rd by order count) risks customer dissatisfaction and could be feeding refund/cancellation behavior on future orders
Short-distance orders cancel more than longer-distance ones	Short: 12.10% cancellation vs Medium 6.69% and Long 7.53%	Counter to expectation; suggests the cancellation driver for short-distance orders is not delivery logistics but something else (e.g., order accuracy, speed expectations, or duplicate ordering) – Insufficient evidence to identify the specific cause from this data
Customer record count mismatch	Segment table shows 1,000 total customers (556+295+149), but customer/order-level tables (Customer Order Frequency, Revenue by Customer, etc.) list only 980 unique customer IDs	Roughly 20 customers appear in the segmentation base but have no recorded orders in the order-level tables – a data completeness/join issue worth resolving before further segment-based revenue-per-customer calculations are trusted
Refund-prone kitchens differ entirely from cancellation-prone kitchens	Top cancellation kitchens (KIT-017, KIT-009, KIT-001, KIT-005) do not appear in the top refund-rate kitchens (KIT-019, KIT-010, KIT-008, KIT-013)	Confirms cancellation and refund issues need separate root-cause investigation – a single “kitchen quality” narrative doesn’t fit both problems

5. Business Opportunities

Opportunity	Supporting Evidence	Suggested Business Action
Target the four worst-cancellation kitchens specifically	KIT-017, KIT-009, KIT-001, KIT-005 combined represent 3,788 orders and a cancellation rate roughly 1.7–1.9x the network average	Run a focused operational review (staffing, order acceptance process, inventory) at these four kitchens before any network-wide cancellation initiative
Re-examine pricing on high-volume, low-revenue categories	Beverage and Dessert have order counts near the category average (3,983 and 4,127 respectively) but the two lowest average unit prices (₹95.79 and ₹130.25) and lowest total revenue (₹837,752 and ₹1,153,364)	Evaluate whether beverage/dessert pricing or bundling could lift revenue without needing to grow order volume, since demand for these categories is already comparable to higher-revenue categories

Opportunity	Supporting Evidence	Suggested Business Action
Reinforce weekend and evening capacity	Weekend order density is higher per day than weekday (6,589 weekend orders over 2 days/week vs 13,411 weekday orders over 5 days/week), and the 18:00–21:00 window alone carries 33.7% of daily order volume	Align staffing schedules and kitchen resourcing specifically to the weekend and evening peak rather than a flat daily allocation
Investigate the New-customer cancellation gap	New segment cancellation rate (11.14%) is the highest of the three segments, while New customers already have the lowest average spend (₹6,597.31)	A first-order-experience or onboarding review could reduce early cancellations and improve conversion of New customers into repeat (Regular/Premium) customers
Reconcile the 20-customer data gap before deeper segment analysis	1,000 customers counted in segmentation vs 980 with recorded orders	Fix the underlying data/join issue so future customer-level revenue and frequency metrics are reliable

6. Final Business Takeaways

1. The business's core financial leakage is concrete and measurable: **₹1.64M (14.3% of total revenue) is tied up in cancelled or refunded orders**, and the leakage is not evenly spread – it's concentrated in four specific kitchens.
2. **Menu revenue is a pricing story, not a demand story** – every category sells in roughly the same volume, so Pizza's revenue lead comes from its price point, not popularity, and lower-revenue categories have room to be repriced rather than needing volume growth.
3. **The business is not dependent on any single customer or kitchen for survival**, but it is heavily reliant on one kitchen (KIT-002) for scale – any disruption there would have an outsized effect on total order volume and revenue.
4. **Cancellation and refund problems are two distinct issues** with different kitchen “offenders” and different customer segments most affected (New customers cancel more, Premium customers refund more) – they need separate diagnostic work, not a single fix.
5. **Demand is highly predictable in time**: it peaks hard between 18:00–21:00 and on week-ends. This is the single most reliable pattern in the entire dataset and should directly inform staffing and kitchen-capacity decisions.

Note: All figures above are drawn directly from the provided SQL query result exports. Where the data did not support a conclusion, this is explicitly marked as “Insufficient evidence.”